

Electronic devices and Circuit

Introduction to Semiconductors:

Semiconductors: Semiconductors are materials which have a conductivity between conductors (generally metals) and nonconductors or insulators (such as most ceramics). Semiconductors can be pure elements, such as silicon or germanium, or compounds such as gallium arsenide or cadmium selenide.

Properties of Semiconductor:

1. The resistivity of a semiconductor is less than an insulator but higher than a conductor.
2. Semiconductors show a negative temperature coefficient of resistance. In simple words, the resistance of the semiconductors decreases as the temperature increases and vice-versa.
3. At zero kelvin, semiconductors behave as insulators. As the temperature is increased it works as a conductor
4. The conductivity of the semiconductors increases when impurities are added. The process of adding impurities to semiconductors is called doping.
5. The vacancy for an electron in the valence band is called a hole in a semiconductor. It acts as a positive charge of magnitude $+e$.

Energy Band Theory

According to Bohr's theory, every shell of an atom contains a discrete amount of energy at different levels. Energy band theory explains the interaction of electrons between the outermost shell and the innermost shell. Based on the energy band theory, there are three different energy bands:

1. Valence band
2. Forbidden energy gap
3. Conduction band

Valence Band

The electrons in the outermost shell are known as valence electrons. These valence electrons contain a series of energy levels and form an energy band known as the valence band. The valence band has the highest occupied energy.

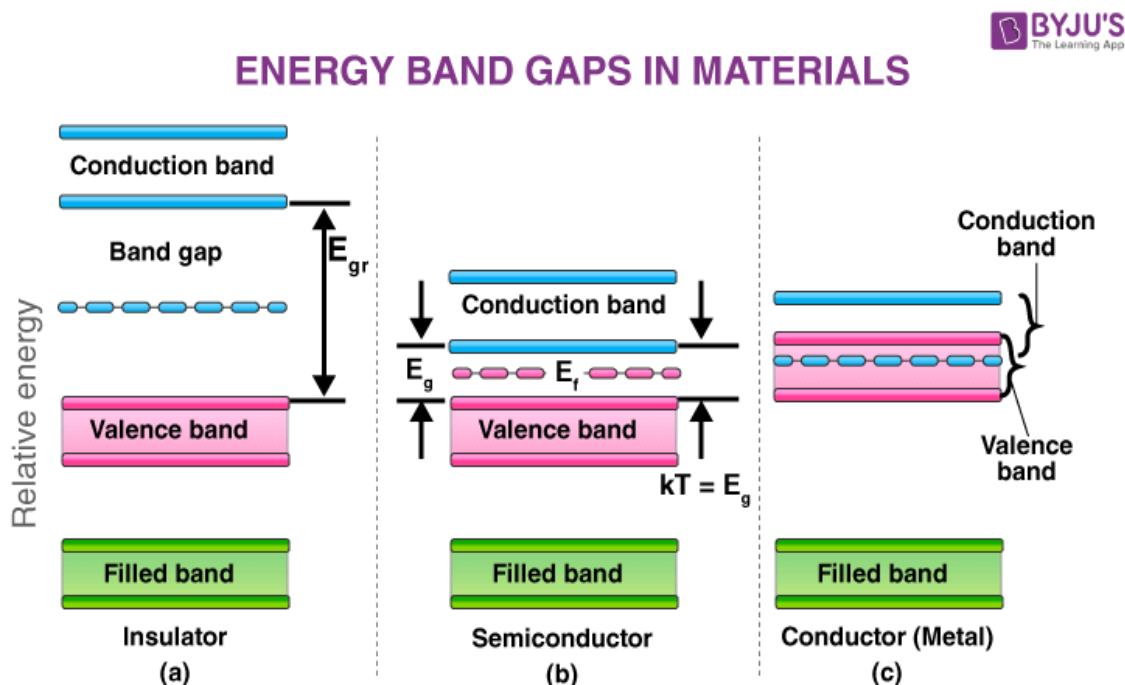
Conduction Band

The valence electrons are not tightly held to the nucleus due to which a few of these valence electrons leave the outermost orbit even at room temperature and become free electrons. **The free electrons conduct current in conductors and are therefore known as conduction electrons.** The conduction band is one that contains conduction electrons and has the lowest occupied energy levels.

Forbidden Energy Gap

The gap between the valence band and the conduction band is referred to as the forbidden gap. As the name suggests, the forbidden gap doesn't have any energy and no electrons stay in this band. If the forbidden energy gap is greater, then the valence band electrons are tightly bound or firmly attached to the nucleus. We require some amount of external energy that is equal to the forbidden energy gap.

The figure below shows the conduction band, valence band and the forbidden energy gap.



Classification of material according to energy band theory:

Conductors

Gold, Aluminum, Silver, Copper, all these metals allow an electric current to flow through them. **There is no forbidden gap between the valence band and conduction band which results in the overlapping of both the bands.** The number of free electrons available at room temperature is large.

Insulators

Glass and wood are examples of the insulator. These substances do not allow electricity to pass through them. They have high resistivity and very low conductivity.

The energy gap in the insulator is very high up to 7eV. The material cannot conduct because the movement of the electrons from the valence band to the conduction band is not possible.

Semiconductors

Germanium and Silicon are the most preferable material whose electrical properties lie in between semiconductors and insulators. The energy band diagram of semiconductors is shown **where the conduction band is empty and the valence band is completely filled but the forbidden gap between the two bands is very small that is about 1eV.** For Germanium, the forbidden gap is 0.72eV and for Silicon, it is 1.1eV. Thus, semiconductor requires small conductivity.

Classification of semiconductor

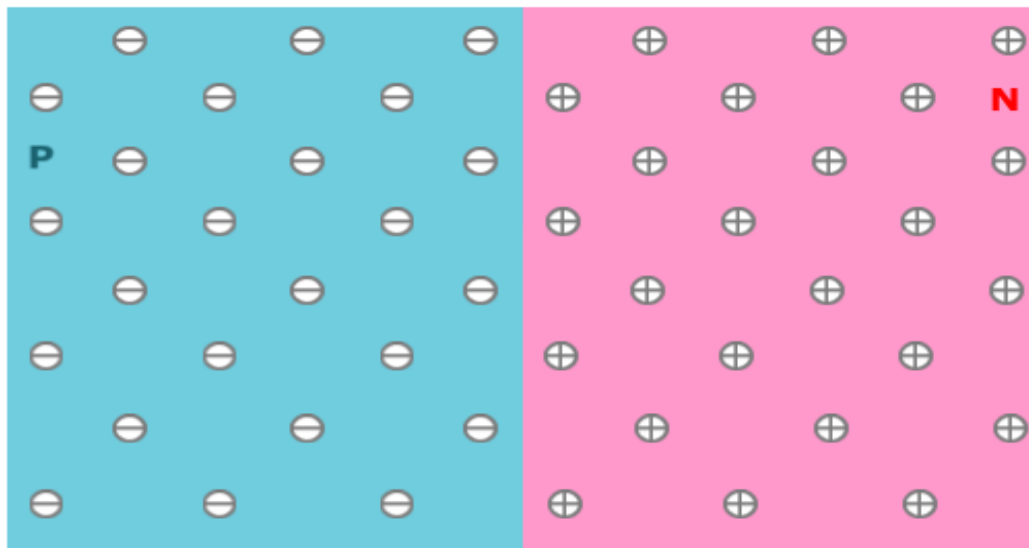
1.Intrinsic Semiconductors: When impurity does not add to the pure semiconductor the it is said to be intrinsic semiconductor or pure semiconductor.

2.Extrinsic Semiconductors: When impurity is added to the pure semiconductor the it is said to be intrinsic semiconductor.

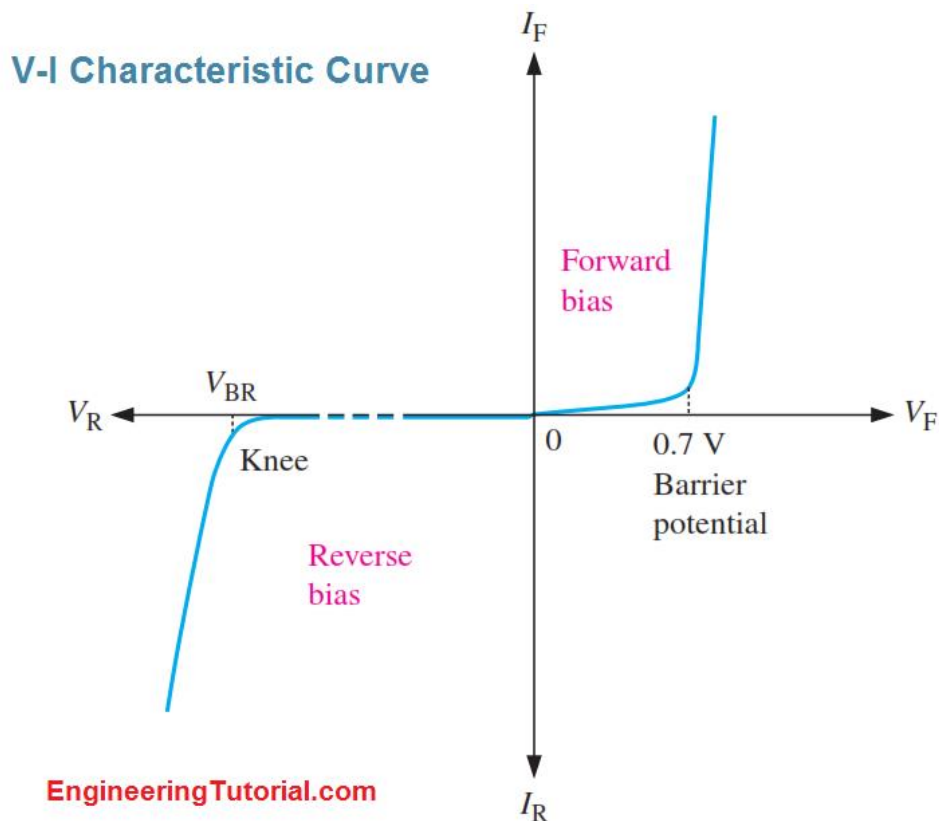
Semiconductor Diodes and Special Purpose Diodes:

Formation of PN-junction

P-n junctions are formed by joining *n*-type and *p*-type semiconductor materials, as shown below. Since the *n*-type region has a high electron concentration and the *p*-type a high hole concentration, electrons diffuse from the *n*-type side to the *p*-type side. Similarly, holes flow by diffusion from the *p*-type side to the *n*-type side. If the electrons and holes were not charged, this diffusion process would continue until the concentration of electrons and holes on the two sides were the same, as happens if two gasses come into contact with each other. However, in a *p-n* junction, when the electrons and holes move to the other side of the junction, they leave behind exposed charges on dopant atom sites, which are fixed in the crystal lattice and are unable to move. On the *n*-type side, positive ion cores are exposed. On the *p*-type side, negative ion cores are exposed. An electric field E forms between the positive ion cores in the *n*-type material and negative ion cores in the *p*-type material. This region is called the "depletion region" since the electric field quickly sweeps free carriers out, hence the region is depleted of free carriers. A "built-in" potential V_{bi} is formed at the junction due to E . The animation below shows the formation of the E at the junction between *n* and *p*-type material.



Properties and V-I Characteristics:



- In forward bias condition p-type is connected to positive terminal of battery and the n-type to the negative terminal of the battery, there is a reduction in the potential barrier, in this condition. For germanium diodes, when the voltage is 0.3 V, and for silicone diodes, when the voltage is 0.7 V the potential barriers decrease and there is a flow of current.
- When the diode is **in forward bias**, as the voltage applied to the diode is overcoming the potential barrier, the current increases slowly and the curve obtained is non-linear. Once the potential barrier is crossed by the diode, the diode behaves normally and the curve rises sharply as further external voltage increases and the curve obtained is linear.
- When the PN junction diode is under **reverse bias**, this results in an increase in the potential barrier and resistance also increases. Minority carriers are present in the junction which creates reverse saturation current flows in the beginning.
- If the applied voltage increases rapidly, there is increased kinetic energy due to minority charge carriers which affect the majority charges. In this stage the diode breaks down. or the voltage is called breakdown voltage, this may also destroy the diode.

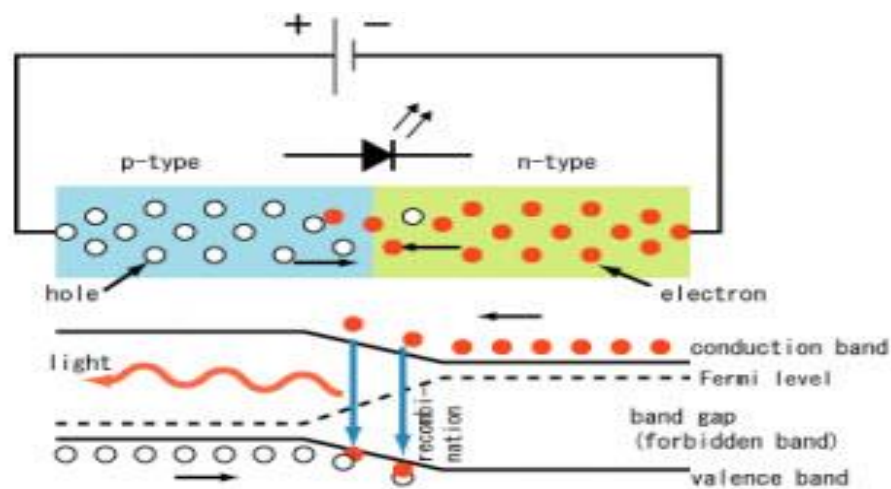
Operations and Uses of Special Diode with Applications

Light-emitting Diode (LED)

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it.

Working Principle of LED

The working principle of the Light-emitting diode is based on the quantum theory. The quantum theory says that when the electron comes down from the higher energy level to the lower energy level then, the energy emits from the photon. The photon energy is equal to the energy gap between these two energy levels. If the PN-junction diode is in the forward biased, then the current flows through the diode.



The flow of current in the semiconductors is caused by the flow of holes in the opposite direction of current and the flow of electrons in the direction of the current. Hence there will be recombination due to the flow of these charge carriers.

The recombination indicates that the electrons in the conduction band jump down to the valence band. When the electrons jump from one band to another band the electrons will emit the electromagnetic energy in the form of photons and the photon energy is equal to the forbidden energy gap.

For example, let us consider the quantum theory, the energy of the photon is the product of both the Planck constant and frequency of electromagnetic radiation. The mathematical equation is shown

$$E_q = hf$$

Where h is known as a Planck constant, and the velocity of electromagnetic radiation is equal to the speed of light i.e c . The frequency radiation is related to the velocity of light as $f = c / \lambda$. λ is denoted as a wavelength of electromagnetic radiation and the above equation will become as a

$$E_q = hc / \lambda$$

From the above equation, we can say that the wavelength of electromagnetic radiation is inversely proportional to the forbidden gap. In general silicon, germanium semiconductors this forbidden energy gap is between the conduction and valence bands are such that the total radiation of electromagnetic wave during recombination is in the form of infrared radiation. We can't see the wavelength of infrared because they are out of our visible range.

The infrared radiation is said to be as heat because the silicon and the germanium semiconductors are not direct gap semiconductors rather these are indirect gap semiconductors. But in the direct gap semiconductors, the maximum energy level of the valence band and minimum energy level of the conduction band does not occur at the same moment of electrons. Therefore, during the recombination of electrons and holes are migration of electrons from the conduction band to the valence band the momentum of the electron band will be changed.

Application of Light Emitting Diode

LEDs used for TV Backlighting

A TV's backlight is the major power-consuming source. Uses of LEDs can give an efficient power reduction. on the edges of the TV, using an LED will be a cost reduction choice. Using LEDs directly behind the display provides better contrast. When it comes to TV backlighting, LEDs have taken the place of CFLs and LCDs.

It is used for Smartphone Backlighting

The backlight design of the smartphone may be slimmer and less expensive thanks to the usage of LED. The price of LEDs may vary according to the size of the smartphone display. They provide greater battery life due to the lower output voltage.

Uses of LED in Displays

LED display boards are common these days and are used outdoors like storage signs, billboards, road signs, etc. In sign boards, which have multiple languages conveying signals, the use of more LEDs will be beneficial in terms of less power consumption.

These are used in Automotives

The Use of LEDs in the automotive industry is growing. With LEDs, energy is saved and there is clearer visibility. These are extensively used in the back and rear of an automobile for better accessibility. LED lighting can improve the safety of pedestrians and drivers as it enhances visibility when it is ON, OFF, and dimmed in any part of the journey.

LEDs used in Dimming of Lights

Few LED applications include dimming of lights which helps in reducing energy consumption. This dimming feature is also used in Appliances where it is of two types. Global Dimming where all LEDs are dimmed together. Local Dimming where LEDs are dimmed independently.

Photo diode

A photodiode is a PN-junction diode that consumes light energy to produce an electric current. They are also called a photo-detector, a light detector, and a photo-sensor. Photodiodes are designed to work in reverse bias condition. Typical photodiode materials are Silicon, Germanium and Indium gallium arsenide.

Working principle of Photodiode

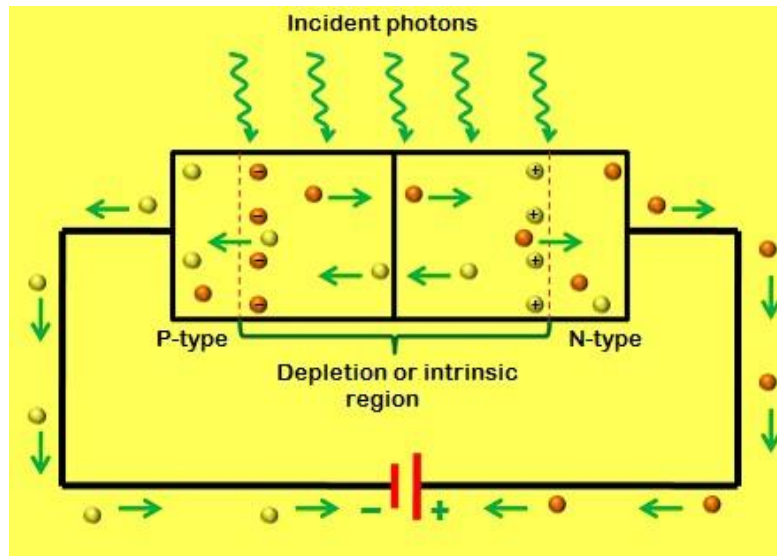


Figure: Photodiode

In general, an electron will have a negative charge and holes will have a positive charge. The depletion energy will have built-in electric field. Due to that electric field, electron-hole pairs move away from the junction. Hence, holes move to anode and electrons move to the cathode to produce photocurrent.

The photon absorption intensity and photon energy are directly proportional to each other. When energy of photos is less, the absorption will be more. This entire process is known as Inner Photoelectric Effect.

Intrinsic Excitations and Extrinsic Excitations are the two methods via which the photon excitation happens. The process of intrinsic excitation happens, when an electron in the valence band is excited by photon to conduction band.

Application of Photodiode

- Photodiodes are used in simple day-to-day applications. The reason for their prominent use is their linear response of photodiode to light illumination.

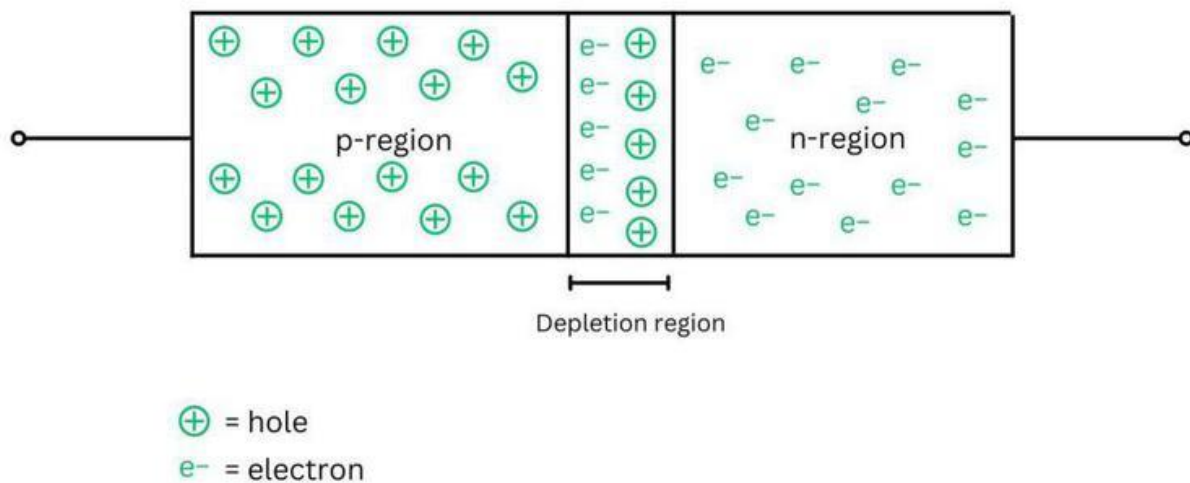
- Photodiodes with the help of optocouplers provide electric isolation. When two isolated circuits are illuminated by light, optocouplers are used to couple the circuit optically. Optocouplers are faster compared to conventional devices.
- Photodiodes are used in safety electronics such as fire and smoke detectors.
- Photodiodes are used in numerous medical applications. They are used in instruments that analyze samples, detectors for computed tomography and also used in blood gas monitors.
- Photodiodes are used in solar cell panels.
- Photodiodes are used in logic circuits.
- Photodiodes are used in the detection circuits.
- Photodiodes are used in character recognition circuits.
- Photodiodes are used for the exact measurement of the intensity of light in science and industry.
- Photodiodes are faster and more complex than normal PN junction diodes and hence are frequently used for lighting regulation and optical communication.

Tunnel Diode

A tunnel diode is a very heavily doped p-n junction diode. In a Tunnel diode electric current decreases as the applied voltage increases, and at high voltage, it works as an ordinary p-n junction diode. In the Tunnel Diode, electric current is due to the “Tunneling effect”. Due to its fast switching ability, it is used in high-frequency oscillators and amplifiers. Due to the tunneling effect, charge carriers move across the depletion layer at very low voltage. The circuit shown below is the symbolic diagram of the Tunnel diode. In tunnel diode the p-terminal acts as an anode and the n-terminal acts as a cathode.

Working principle of Tunnel diode

Width of the depletion region depends on the level of doping in semiconductor materials, If the doping level is high then the width of the depletion region is thin and if the doping level is low then width of depletion layer is more. In tunnel diode the width of the depletion region is very thin (because tunnel diode is very highly doped semiconductor device), thus when low voltage is applied across the terminal in forward biasing electron moves across the depletion from n-type semiconductor towards p-type semiconductor.



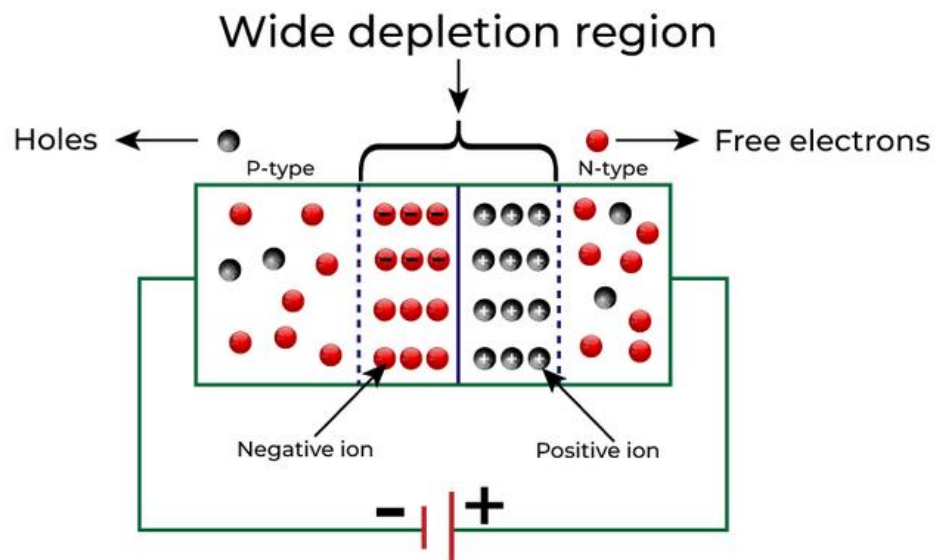
Application of tunnel diode

- Tunnel diode can be used as a switch, amplifier, and oscillator.
- Since it shows a fast response, it is used as high frequency component.
- Tunnel diode acts as logic memory storage device.
- They are used in oscillator circuits, and in FM receivers. Since it is a low current device, it is not used more.

Varactor Diode

A varactor diode, also known as a Varicap or volt-cap, is a type of PN junction diode primarily utilized in the reverse-biased mode. It is a device whose capacitance varies with the variation in the applied reverse bias potential. The term “Varicap” originates from the fusion of the word’s “variable” and “capacitor.”

Working principle of Varactor Diode



When a diode is subjected to forward bias, the majority carriers in both the p and n regions are pushed in the direction of the battery’s applied voltage, leading to a reduction in the width of the depletion region. Eventually, the diode begins to conduct current more readily. However, a varactor diode is designed with a specific focus on its capacitance properties and its ability to store charges. This is why it is typically operated in reverse bias. When reverse biased, the varactor diode’s capacitance varies with changes in the applied reverse voltage, making it useful in various applications like tuning circuits, frequency modulation, and voltage-controlled oscillators.

Application of Varactor Diode

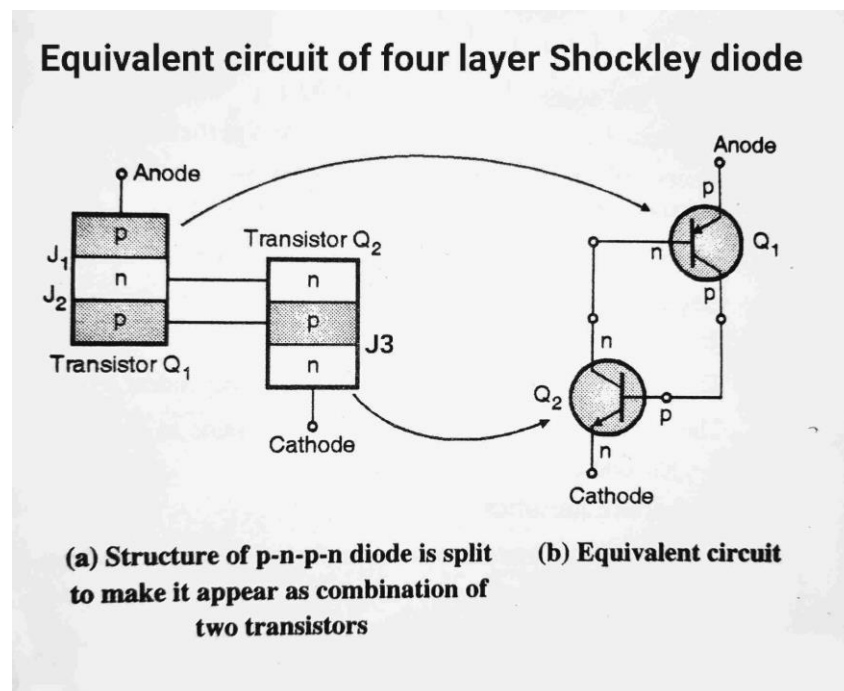
- **Voltage-Controlled Oscillators (VCOs):** Varactor Diodes are used in VCOs. VCOs are used in Phase-locked Loops and Communication Systems. By Varying the Voltage Across the Varactor Diode, the Capacitance Changes which leads to change of the Frequency in the Oscillator.

- **Frequency Modulation Tuning in Radios:** The Varactor Diodes are also used in the RF Circuits. By Varying the Voltage Across the Varactor Diode, we can adjust the resonance Frequency of the tuned Circuit which we allow the tuning in FM radios.
- **Phase Shifters:** Varactor Diodes are also Implemented in Phase array Antennas. By Varying the Capacitance in the Certain part of the Antenna Array can lead to Controlling of the phase of the Emitted Signal Which can be used for beam Steering and Shaping.
- **Voltage-Controlled Filters:** Varactor Diodes are used in Voltage Controlled Bandpass or lowpass Filters. The Cutoff Frequency of the Filter can be Adjusted by Varying the Voltage across the varactor diode.

Shockley Diode

A Shockley diode is a four-layer semiconductor diode made from alternating layers of P-type and N-type semiconductor material to form a PNPN structure. It is also known as a four-layer diode or PNPN diode.

Working Principle of Shockley Diode



a.) Operation in forward biased:

Due to the four alternate layers, there are three junctions in the Diode. When an external voltage is applied to make the anode positive with respect to cathode, the junctions J1 and J3 are forward biased and junction J2 is reverse biased. The externally applied voltage appears almost completely across the reverse biased junction J2. A very small current (leakage current) will flow through the diode. As we increase the external voltage, the diode current increases slowly. This will continue upto a voltage called “firing voltage” or “breakover voltage” V_{bo} . When the current increases to a particular value the diode switches to ON state. This value of current is called holding current in Shockley diode.

As soon as the applied voltage becomes equal to the breakover voltage, the diode current increases sharply and the voltage across the device decreases sharply. This happens due to breakdown of junction J2 at the breakover voltage. Thus at the breakover point the p-n-p-n diode switches from OFF state to ON state. This is shown in the VI characteristics.

b.) Operation when reverse biased:

When reverse biased the p-n-p-n diode acts like a typical p-n junction diode. Junction J1 and J3 are reverse biased and junction J2 is forward biased. The current flowing through the device is very small leakage current. If the reverse voltage reaches the reverse breakdown voltage then the device breakdown and a large current flow through it and diode gets damaged.

Application of Shockley Diode

- **Voltage Regulators:** The Shockley diode can be used as a voltage regulator, ensuring that the output voltage remains constant regardless of fluctuations in the input voltage. This is particularly useful in power supply circuits, where maintaining a stable output voltage is crucial.
- **Overvoltage Protection:** The device can act as a protective element in circuits by clamping the voltage to a specified level, preventing damage to sensitive components due to overvoltage conditions.

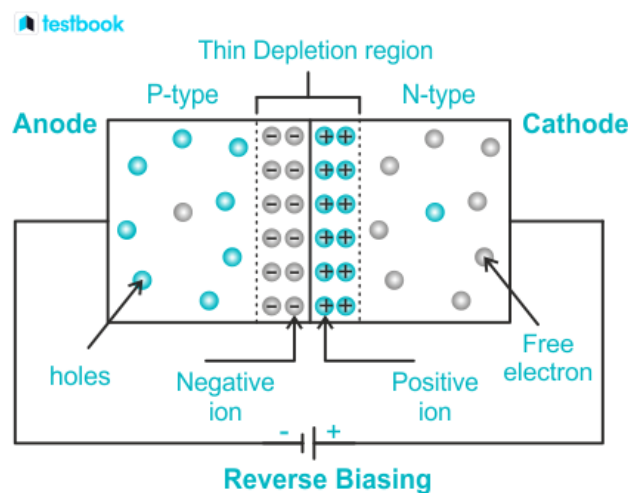
- **Switching Applications:** The Shockley diode's ability to switch between high and low impedance states makes it suitable for various switching applications, such as triggering a relay or controlling the firing angle of a thyristor in a phase-controlled rectifier.

Zener Diode

A Zener diode is a highly doped semiconductor device specifically designed to function in the reverse direction. It is engineered with a wide range of Zener voltages (V_z), and certain types are even adjustable to achieve variable voltage regulation.

Working Principle of Zener Diode

The working principle is such that if the reverse bias voltage is less than the breakdown voltage, or if it is forward biased then it acts as an ordinary diode. This means that forward bias allows current to flow and reverse bias blocks the current from flowing. After this, the voltage surpasses the breakdown point in reverse bias, and the diode falls in the Zener region, where it gets conducted without getting damaged. Current in this region is known as avalanche current but for a Zener diode, it is also known as a Zener current.



When the voltage decreases in the circuit the diode maintains its non-conducting condition and gets back to its natural properties. This specific property of the Zener diode of being functional in the reverse bias and with avalanche current is given by the rich doping of the semiconductor material present in it.

Further, by controlling the amount of doping of the semiconductor material and by doing this the thickness of the depletion region in the PN junction and the breakdown voltage can be set to any value according to the need of the appliance.

Application of Zener Diode

Zener diode as a voltage regulator:

The Zener diode is used as a Shunt voltage regulator for regulating voltage across small loads. The Zener diode is connected parallel to the load to make it reverse bias, and once the Zener diode exceeds knee voltage, the voltage across the load will become constant. The breakdown voltage of Zener diodes will be constant for a wide range of currents.

Zener diode in over-voltage protection:

When the input voltage is higher than the Zener breakage voltage, the voltage across the resistor drops resulting in a short circuit, this can be avoided by using the Zener diode.

Zener diode in clipping circuits:

Zener diode is used for modifying AC waveform clipping circuits by limiting the parts of either one or both the half cycles of an AC waveform.

Bipolar Junction Transistors

Transistors:

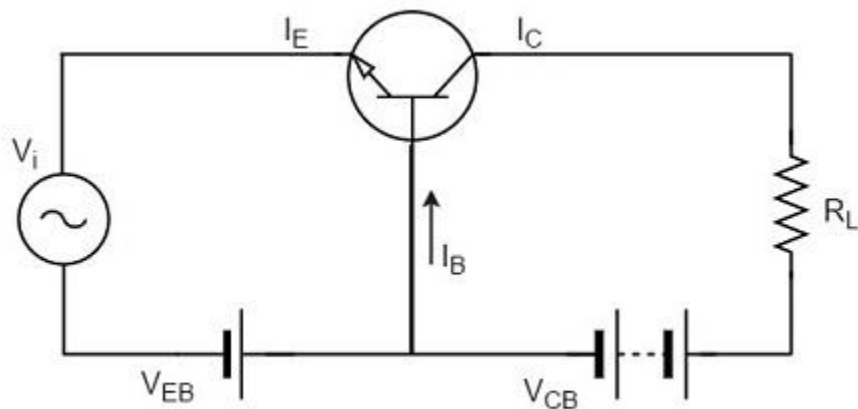
A transistor is a semiconductor device used to amplify or switch electronic signals.

Transistors are broadly divided into three types:

1. Bipolar transistors (bipolar junction transistors: BJTs),
2. Field-effect transistors (FETs),
3. Insulated-gate bipolar transistors (IGBTs).

Amplifying Actions of Transistor

A transistor acts as an amplifier by raising the strength of a weak signal. The DC bias voltage applied to the emitter base junction, makes it remain in forward biased condition. This forward bias is maintained regardless of the polarity of the signal. The below figure shows how a transistor looks like when connected as an amplifier.



The low resistance in input circuit, lets any small change in input signal to result in an appreciable change in the output. The emitter current caused by the input signal contributes the collector current, which when flows through the load resistor R_L , results in a large voltage drop across it. Thus, a small input voltage results in a large output voltage, which shows that the transistor works as an amplifier.

Switching Actions of Transistor

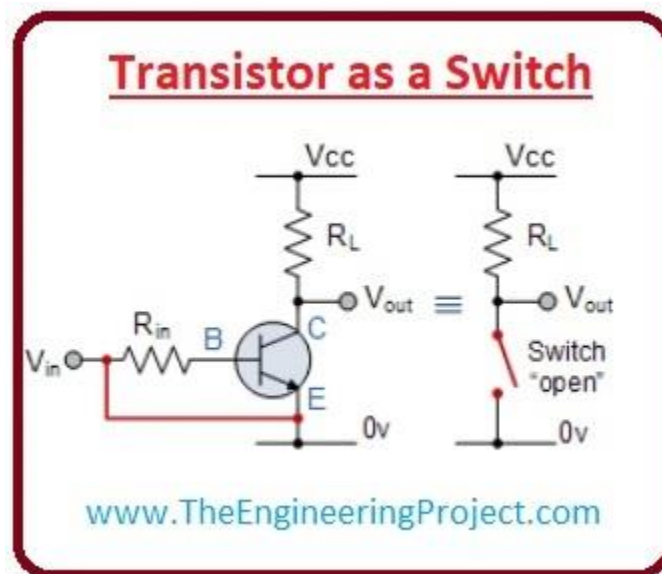
A transistor is a semiconductor device that can amplify or switch electronic signals. When used as a switch, a transistor operates in one of two modes: cutoff or saturation.

In the cutoff mode, the transistor acts as an open switch, and no current flows through it. In the saturation mode, the transistor acts as a closed switch, and a large current can flow through it.

To use a transistor as a switch, it is connected in a specific configuration called a common-emitter configuration. In this configuration, the transistor is connected between the power source and the load.

When a small current is applied to the base of the transistor, it allows a much larger current to flow through the collector and emitter terminals, effectively turning the transistor on. Conversely, when no current is applied to the base, the transistor turns off and the current flow stops.

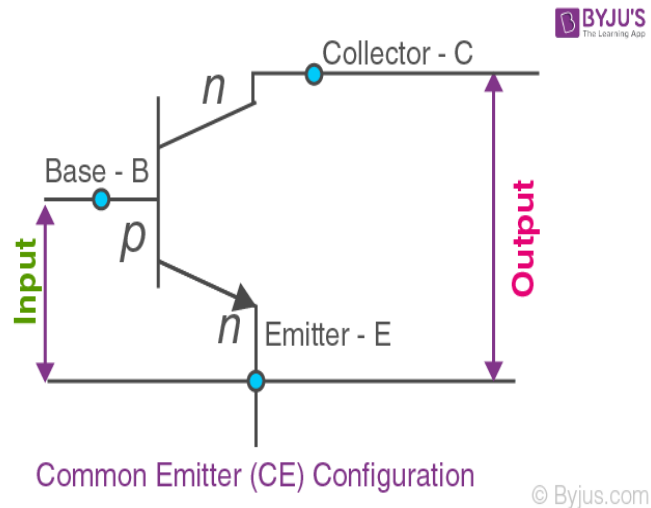
By controlling the current to the base, the transistor can be switched on and off rapidly, making it useful for digital applications such as logic gates, amplifiers, and power control circuits.



Transistor Characteristics In CB, CE and CC Configurations

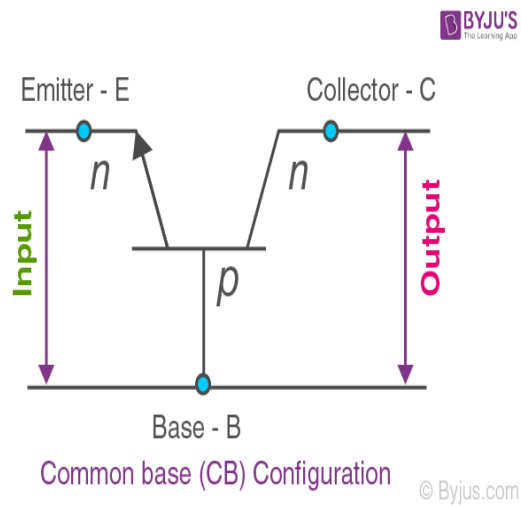
Common Emitter (CE) Configuration of Transistor

The configuration in which the emitter is connected between the collector and base is known as a common emitter configuration.



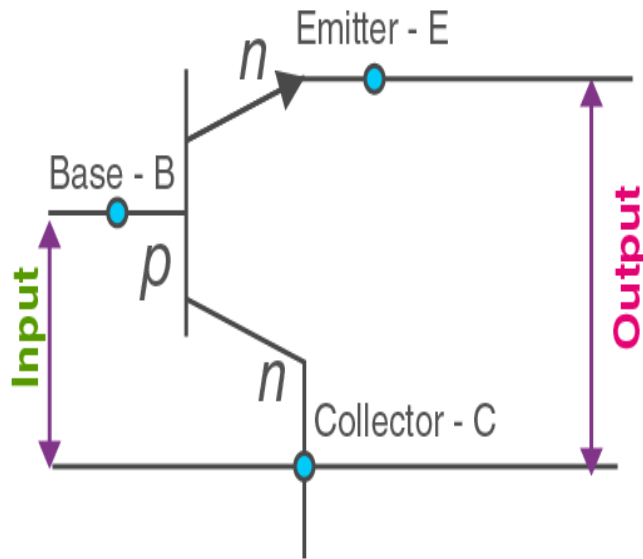
Common Base (CB) Configuration of Transistor

In CB Configuration, the base terminal of the transistor will be connected common between the output and the input terminals.



Common Collector (CC) Configuration of Transistor

In CE Configuration, the Collector terminal of the transistor will be connected common between the output and the input terminals.

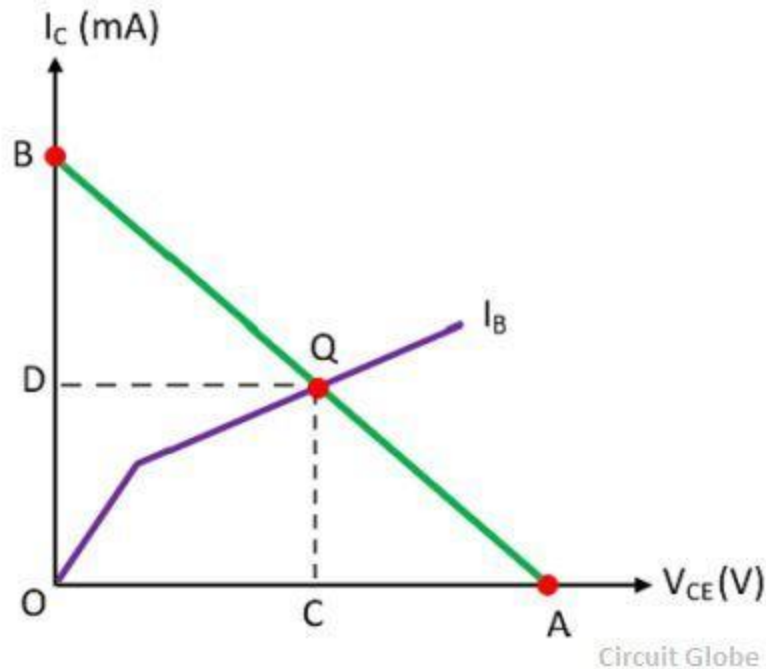


Common Collector (CC) Configuration © Byjus.com

Operating Point of a transistor

The point which is obtained from the values of the I_C (**collector current**) or V_{CE} (**collector-emitter voltage**) when **no signal** is given to the input is known as the **operating point or Q-point** in a transistor. It is called operating point because variations of I_C (collector current) and V_{CE} (collector-emitter voltage) takes place around this point when no signal is applied to the input.

The operating point is also called **quiescent (silent) point** or simply Q-point because it is a point on $I_C - V_{CE}$ characteristic when the **transistor is silent** or no input signal is applied to the circuit. The operating point can be easily obtained by the DC load line method. The DC load line is explained below.



Let, determines the operating point of particular base circuit current I_B . According to the load line condition, the $OA = V_{CE} = V_{CC}$ and $OB = I_C = V_{CC}/R_C$ is shown on the output characteristic curve above. The point Q is the operating point where the DC load line intersects the base current I_B at the output characteristic curves in the absence of input signal.

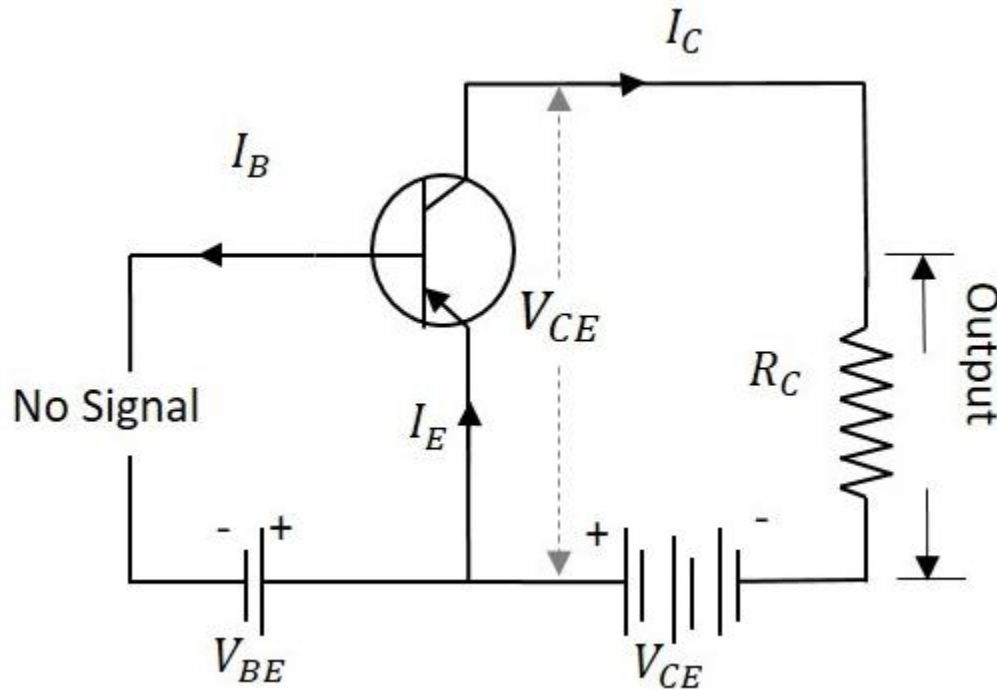
Where $I_C = OD$ mA

$V_{CE} = OC$ volts.

The position of the Q-point depends on the applications of the transistor. If the transistor is used as a switch then for open switch the Q-point is in the cutoff region, and for the close switch, the Q-point is in the saturation region. The Q-point lies in the middle of the line for the transistor which operates as an amplifier.

Transistor Load Line Analysis

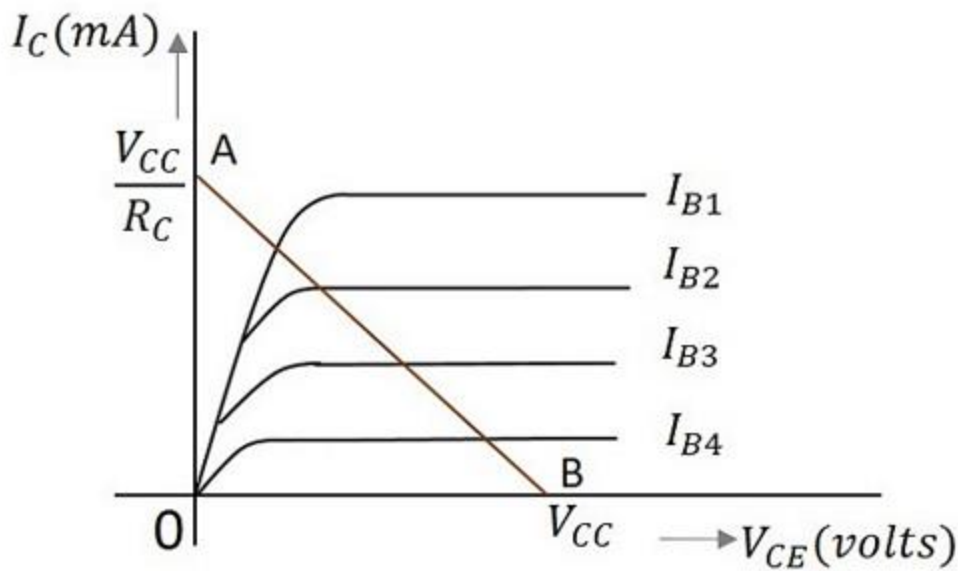
When the transistor is given the bias and no signal is applied at its input, the load line drawn at such condition, can be understood as **DC** condition. Here there will be no amplification as the signal is absent. The circuit will be as shown below.



The value of collector emitter voltage at any given time will be

$$V_{CE} = V_{CC} - I_C R_C$$

As V_{CC} and R_C are fixed values, the above one is a first-degree equation and hence will be a straight line on the output characteristics. This line is called as **D.C. Load line**. The figure below shows the DC load line.



To obtain the load line, the two end points of the straight line are to be determined. Let those two points be A and B.

To obtain A

When collector emitter voltage $V_{CE} = 0$, the collector current is maximum and is equal to V_{CC}/R_C . This gives the maximum value of V_{CE} . This is shown as

$$V_{CE} = V_{CC} - I_C R_C$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C}$$

This gives the point A ($OA = V_{CC}/R_C$) on collector current axis, shown in the above figure.

To obtain B

When the collector current $I_C = 0$, then collector emitter voltage is maximum and will be equal to the V_{CC} . This gives the maximum value of I_C . This is shown as

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = V_{CC} \quad (\text{As } I_C = 0)$$

This gives the point B, which means ($OB = V_{CC}$) on the collector emitter voltage axis shown in the above figure.

Hence, we got both the saturation and cut-off point determined and learnt that the load line is a straight line. So, a DC load line can be drawn.